

Substance abuse related disorder

Introduction

- Substance use disorder is a chronic disease with progressive **deterioration of psychological and physiologic** activity secondary to the habitual use of a drug.
- This complex disease disrupts many aspects of an individual's life

Introduction

- ❑ **Addiction:** A primary chronic neurobiologic disease, with genetic, psychosocial, and environmental factors influencing its development and manifestations
- ❑ it is characterized by one or more of the following five Cs:
 - Chronicity
 - impaired control over drug use
 - compulsive use
 - continued use despite harm, and
 - craving.

Introduction...

- **Intoxication:** Development of a substance-specific syndrome after recent ingestion
- **Physical dependence:** A state of adaptation manifested by a drug class–specific withdrawal syndrome that can be produced by abrupt cessation, rapid dose reduction, decreasing blood level of the drug, and/or administration of an antagonist
- **Substance dependence:** continued maladaptive pattern of substance use in spite of repeated adverse consequences related to the repeated use.

Introduction...

- **Substance abuse:** A maladaptive pattern of substance use characterized by repeated adverse consequences related to the repeated use of the substance
- **Tolerance:** A state of adaptation in which exposure to a drug induces changes that result in a diminution of one or more of the drug's effects over time
- **Withdrawal:** The development of a substance-specific syndrome after cessation of or reduction in intake of a substance that was used regularly.

1. Alcohol

- Alcohol use disorder can result in **toxicity, withdrawal, and addiction**
- Alcohol toxicity is an acute, life-threatening condition that requires aggressive medical attention
- Alcohol withdrawal is the neurobiological syndrome associated with increased tolerance or physical dependence resulting from chronic alcohol consumption
- Acute detoxification and withdrawal after chronic alcohol abuse is a serious condition that can require hospitalization and adjunctive pharmacotherapy

- Alcohol dependence is a life-long relapsing disorder that consists of signs of alcohol abuse
- Signs of Alcohol abuse include: **continued drinking** despite alcohol-related physical, social, psychological, or occupational problems, or **drinking in dangerous situations** to the extent that the person also experiences at least three of the following seven symptoms:
 - Neglect of other activities
 - Excessive use of alcohol
 - Impaired control of alcohol consumption
 - Persistence of alcohol use
 - Large amounts of time spent in alcohol-related activities
 - Withdrawal symptoms
 - Tolerance of alcohol

Epidemiology

In US

- **52.7%** aged 12 or older reported being current drinkers of alcohol (at least one drink in the past 30 days)
- 22.9% participated in binge drinking
- **6.3%** reported **heavy drinking**(five or more drinks on the same occasion on each of 5 or more days in the past 30 days)

In Ethiopia

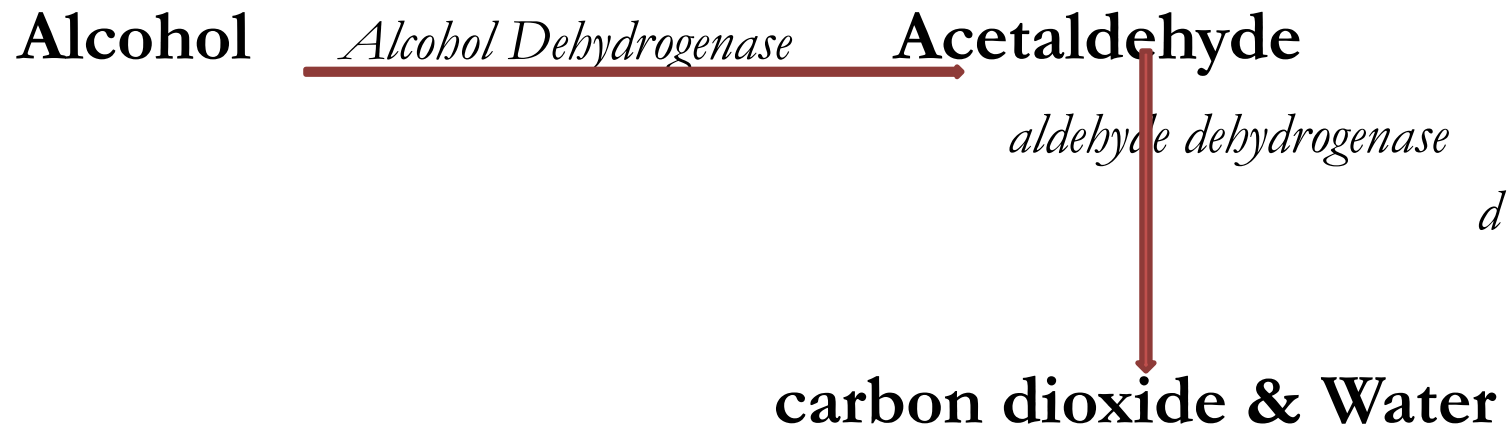
- **23.9%** current user
- **44.2%** lifetime user
- **8.9%** hazardous consumption

Risk of Alcohol Consumption

- CVD
- Hepatic cirrhosis
- GI & Upper airway Cancers
- Fetal abnormalities
- Injuries
- Auto collision
- Violence
- worker productivity and absenteeism
- Affect family interactions and school performance

Pathophysiology

- Alcohol absorption begins in the stomach within 5–10 min of ingestion
- Peak concentrations usually achieved 30–90 min after finishing last drink.



Pathophysiology...

- 12 oz of beer(355ml of 5% beer)= 5 oz (148ml) of 12% wine= 1.5 oz(44ml or one shot) of 40%(80-proof) whiskey and all contain 14 g of alcohol ,
- This amount increases BAC by 20–25 mg/dL in a healthy 70-kg man.
- Deaths from alcohol intoxication generally occur when BACs >500 mg/dL

Clinical presentation
Signs and symptoms of alcohol intoxication:

- Slurred speech
- Ataxia
- Incoordination
- Sedation
- Nystagmus
- Impaired judgment
- Unconsciousness
- Nausea
- Vomiting
- Respiratory depression
- Coma

Clinical presentation...

Signs and symptoms of alcohol withdrawal

- Tachycardia
- Diaphoresis
- Hyperthermia
- Hallucinations
- Delirium
- Seizures

Treatment

Goal of treatment

- ❑ Prevention and treatment of withdrawal symptoms
- ❑ Long-term abstinence after detoxification
- ❑ Entry into ongoing medical and alcohol dependence treatment
- ❑ Medical or psychiatric complications prevention and treatment

Pharmacologic treatment

For intoxication is supportive:

- Respiratory depression- Endotracheal intubation
- CNS depression-Gastric lavage, naloxone, dialysis
- Hypotension-IV fluid
- Hypoglycemia- 50%glucose

Pharmacologic Treatment

For Alcohol withdrawal:

- Benzodiazepines are the current standard of care
- Chlordiazepoxide, diazepam, lorazepam and oxazepam represent the most efficacious pharmacotherapies
- **Lorazepam is preferred BZD**
- Alternative treatment for withdrawal include: Carbamazepine, baclofen
- They are effective in preventing seizure
- untreated withdrawal can result in death.

Pharmacologic treatment...

Adjunctive treatment includes:

- Multivitamin, Thiamine, Crystalloid fluids (DNS+KCL)
- Clonidine for Autonomic tone rebound and Hyperactivity
- Labetalol for hypertensive emergencies
- Antipsychotics, Haloperidol: Agitation & unresponsive to BDZ, hallucinations or delusions

Monitoring of vital signs and fluid and electrolyte levels is essential for severely ill patients with withdrawal

Pharmacologic treatment...

Alcohol dependence:

1. Disulfiram:

- Inhibits aldehyde dehydrogenase, allowing accumulation of acetaldehyde
- Deters drinking by producing aversive reaction if patient drinks
- Wait at least 24 hours after last drink before starting disulfiram

Pharmacologic treatment...

Effect of disulfiram when taken with alcohol include:

- Flushing
- Nausea/Vomiting.
- Headache
- Palpitations
- Tachycardia
- Fever
- Hypotension.

Pharmacologic treatment...

Rare, severe reactions include:

- Respiratory depression
- Arrhythmias
- Congestive Heart Failure
- Myocardial infarction
- Seizures
- Death.

Pharmacologic treatment...

Monitoring:

- Liver function tests at baseline and then 2 weeks, 3 months, 6 months, then twice yearly after starting therapy.

Pharmacologic treatment...

2. Naltrexone:

- Reduce craving and result in fewer drinking days
- CI in patients with hepatitis or liver failure
- Monitor liver function tests monthly for first 3 months, then every 3 months

Side effects include:

- Nausea
- Headache
- Dizziness
- Nervousness
- Insomnia
- Somnolence

Pharmacologic treatment...

3. Acamprosate:

- Reduce craving and increase success in maintaining abstinence
- May be combined with naltrexone and psychosocial intervention.

Most common side effects are GI:

- Diarrhea
- Flatulence
- Upset stomach
- Anorexia

2. Nicotine

- Cigarette smoking is the leading cause of preventable morbidity and mortality

Epidemiology

- 25.5% of the US population 12 years of age and older used a tobacco product at least once a month
- 3.1% in Ethiopia

Pathophysiology

- Nicotine is ganglionic cholinergic-receptor agonist with dose-dependent pharmacologic effects

Effects include:

- CNS and peripheral nervous system stimulation and depression
- Respiratory stimulation
- Skeletal muscle relaxation
- Catecholamine release by adrenal medulla
- Peripheral vasoconstriction

Risks from smoking

It increases the risk of:

- Cardiovascular diseases
- Lung cancer
- Other cancers (eg, bladder, kidney, pancreatic, esophageal, stomach, and head & neck cancers)
- Nonmalignant respiratory diseases
- COPD – 90%

Clinical presentation

Withdrawal symptoms include:

- Anxiety
- Cravings
- Difficulty concentrating
- Frustration
- Irritability
- Hostility
- Insomnia
- Restlessness

Treatment

Goal

- Cessation of nicotine use
- Resolution of withdrawal symptoms
- Avoidance of side effects from pharmacologic therapies

Non pharmacologic treatment

4A's

- Advice to quit
- Assess willingness to quit
- Assist with quitting
- Arrange follow-up

Pharmacologic treatment

1st-line for smoking cessation include:

- Nicotine gum, inhaler, lozenges, nasal spray, or patch
- Bupropion sustained release
- Varenicline
- Consider combinations if single agent failed

2nd-line include:

- Clonidine
- Nortriptyline

Monitoring

- Ask patient about the presence of withdrawal symptoms and nicotine craving
- Monitor for medication side effects at each follow-up visit.
- Monitor blood pressure in patients taking clonidine.
- Screen for psychiatric illness or behavior change after starting varenicline.

thanks...